Data Mining on Imbalanced Data:
Detecting Fraudulent Transactions in Imbalanced Datasets

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Abstract

Fraud detection in financial institutions is a critical activity. Despite the implementation of security measures, fraud continues to spread rapidly. Online and telephone-based payment methods are among the factors contributing to this problem. Therefore, in addition to preventive measures, methods for detecting fraud and preventing fraudulent transactions from being recorded are essential. Financial institutions seek to identify fraudulent activities quickly and efficiently. This paper presents a method intended to achieve good accuracy in detecting fraudulent transactions. Because fraudulent transactions are considerably fewer than legitimate transactions, identifying them is difficult. Such datasets are known as imbalanced datasets. Detecting fraud in this setting requires a simple and effective model that can perform well despite the unequal class distribution. After cleaning the data, we apply two sampling stages: synthetic oversampling of the minority class, followed by simple sampling of the dataset. Finally, we apply the k-nearest neighbours (k-NN) algorithm. The proposed method is reported to perform better than other classifiers, including decision trees and neural networks.

Keywords: data mining; imbalanced datasets; data cleaning; synthetic minority-class oversampling; simple sampling; k-nearest neighbours (k-NN).

1. Introduction

Fraud has existed throughout human history and has taken different forms in different eras. Advances in modern technology have created additional ways to commit fraud. Fraud detection is therefore a vital activity for financial organisations. Banks and credit-card issuers use security measures to prevent the misuse of customers' accounts as far as possible. Nevertheless, the growth of online and telephone purchases and payments, websites distributing malicious software such as password-cracking tools, websites selling users' information, and customers' occasional failure to protect their account details continue to create opportunities for fraud. Consequently, in addition to preventive measures, methods are needed to detect fraud and, where possible, stop it before a transaction is recorded [4].

Since security is a central concern in electronic banking, banks are particularly concerned with ensuring the security of customers' electronic transactions. However, fraudsters often act in sophisticated and intelligent ways. Banks therefore seek to improve their systems so that they can detect fraud and identify suspicious behaviour in online transactions. Credit-card fraud is an important topic in technology research. Various studies identify credit-card transactions as a major area of fraud, and many proposed fraud-detection techniques can also be applied to credit-card transactions. This paper aims to identify fraudulent activity and present a method for detecting it.

1.1. What Is Data Mining?

Data mining can be described in several complementary ways:

1. It is the process of discovering meaningful relationships, patterns, and trends by examining large volumes of data.
2. It involves developing models that provide compact representations of historical data.
3. It summarises relationships in datasets in a form that is understandable and useful to the data owners [3, 4].

1.2. Credit-Card Fraud

Banking fraud takes several forms, one of which is credit-card fraud. In the modern era, fraud is a major cause of financial loss not only for businesses but also for individual customers. As credit-card transactions have increased, credit-card fraud has also become more common in recent years.

Financial institutions have developed two general approaches to identifying credit-card fraud patterns. In the first approach, a bank or financial institution uses data warehouses and data-mining techniques to discover fraud patterns. It then compares those patterns with its own databases to identify possible instances of internal fraud. In the second approach, the bank or financial institution identifies fraud patterns using its own internal information [2, 3].

2. Related Work

Study [5] uses an artificial immune system to detect fraudulent transactions and compares its performance and cost with several other methods, including genetic algorithms, neural networks, naïve Bayes, and decision trees. The proposed method is reported to provide a more optimal result. An artificial immune system is a supervised approach inspired by the human immune system.

Study [1] describes and compares several data-mining methods for fraud detection, including support vector machines (SVM), random forests, and logistic regression. Its dataset was collected over 13 months, from early 2001 to early 2002, and contained 50 million transactions, of which 2,420 were fraudulent. For evaluation, the researchers divided the dataset into four subsets to compare the algorithms under different fraud rates. They used several evaluation metrics. The reported comparisons indicate that random forests achieved the best overall performance.

Study [2] combines clustering and classification, an approach described in the manuscript as semi-supervised learning. After data cleaning and dimensionality reduction using principal component analysis (PCA), the researchers used self-organising maps (SOM) to determine clusters and a k-nearest-neighbour-based method (referred to in the manuscript as K-means) to assign samples to clusters. For classification, they used decision trees, neural networks, and support vector machines. The dataset came from a telecommunications company in Tehran and contained 11,855 transactions, including 8,848 commercial and 2,008 non-commercial transactions. The study compared the methods and reported that its proposed approach achieved the highest performance on this dataset.

Another study discussed in [2] proposes a method that first applies the Synthetic Minority Over-sampling Technique (SMOTE) to the dataset and then performs an additional sampling step using k-nearest neighbours. The resulting balanced datasets are used to train two three-layer feed-forward neural-network classifiers with sigmoid–sigmoid–linear activation functions. Since the final layer is linear, the outputs are decimal values; a threshold is therefore needed to convert them into the two classes, legitimate and fraudulent. A further three-layer neural network with sigmoid–sigmoid–binary separation activation is used to determine an appropriate threshold. The manuscript explains that SMOTE creates synthetic minority-class examples by combining characteristics of existing minority-class samples, thereby increasing the representation of fraudulent transactions. The k-NN step uses Manhattan distance and, in the described experiment, k = 1. The proposed method is compared with SVM, k-NN with different values of k, naïve Bayes, decision trees, and Bayesian networks. The study reports higher accuracy for the proposed method. Its dataset contained 281,104 transactions, of which 281,012 were legitimate and 1,082 were fraudulent.

Study [10] introduces a hybrid model combining k-nearest neighbours, Bayesian methods, and a genetic algorithm. The dataset is classified repeatedly; samples that are incorrectly predicted at each stage are removed, and the final model is built using the selected samples. The model was applied to five datasets, and the results showed a reduction in prediction error. On one dataset, the prediction error rate decreased by 57%. The manuscript notes, however, that the model has limitations, including its lack of suitability for large datasets.

Study [8] uses clustering to investigate samples identified as potentially fraudulent. It examines the behaviour of account holders within a peer group over a period of time. If an account exhibits behaviour that differs significantly from the group, it is flagged as suspicious. In other words, the system analyses an individual cardholder's behaviour over time and identifies unusual patterns for further investigation.

3. Proposed Data-Mining Method

The proposed workflow, illustrated in Figure 1, addresses class imbalance through a sequence of sampling and classification steps. First, SMOTE [1] is used for oversampling, increasing the number of fraudulent samples by generating synthetic examples from the minority class. In the experiment described, the number of fraudulent samples is doubled. Second, simple sampling is used for undersampling [11], reducing the number of legitimate samples. The sample_ratio_per_class parameter controls the proportion of samples retained for each class. In the proposed method, 100% of the fraudulent samples and 90% of the legitimate samples are retained. Third, several algorithms are evaluated to identify the best-performing classifier. The best result is obtained using k-nearest neighbours [10], with k = 3 and Manhattan distance as the distance metric. The k-NN algorithm identifies samples that are most similar to one another according to the selected distance measure.

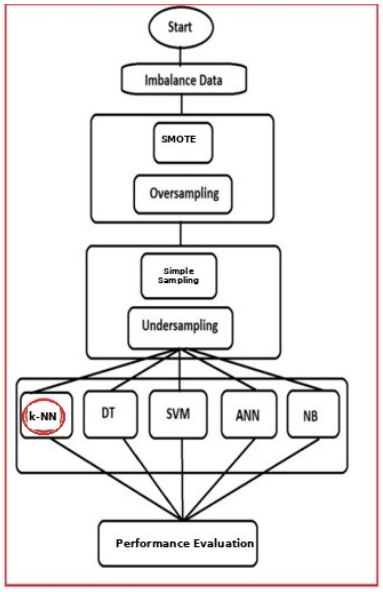


Figure 1. Flowchart of the proposed method.

4. Experiments and Results

4.1. Dataset Description

To implement the proposed method, we used a dataset of fraudulent and legitimate bank transactions obtained from a large Brazilian bank in 2004 [5]. The original manuscript reports 41,647 transactions: 40,088 labelled as legitimate and 1,559 labelled as fraudulent. These counts sum to 41,647. The manuscript reports that 3.74% of the transactions are fraudulent and 96.26% are legitimate, which are consistent with the reported class counts after rounding. The dataset contains 18 features: one target/class variable and 17 predictor variables. It was collected over a three-month period, from July 14, 2004, to September 12, 2004.

4.2. Comparison of the Proposed Method with Other Methods

The proposed method is compared with artificial neural networks (ANN), decision trees (DT) [8], support vector machines (SVM) [10], and naïve Bayes (NB) [11]. Table 1 presents the evaluation results using several metrics, including accuracy, recall, precision, lift, and F-measure.

Method	Accuracy	Recall	Precision	Lift	F-measure
k-NN	97.36%	81.14%	84.96%	1.06800%	83.01%
DT	92.53%	35.86%	54.64%	686.93%	43.13%
NN	93.66%	39.48%	67.27%	845.64%	49.63%
SVM	52.03%	49.71%	8.25%	103.73%	14.15%
NB	96.37%	44.48%	100%	1,528.65%	61.41%

According to Table 1, the proposed classifier achieves the strongest overall balance in identifying legitimate and fraudulent transactions. Although naïve Bayes achieves a precision of 100% and performs closely to the proposed method on some measures, it does not identify legitimate transactions as effectively. The manuscript therefore considers the proposed method the best overall approach for this dataset because it provides a stronger combined result across the two classes.

5. Conclusion

This paper first examined the challenges of banking fraud detection and then proposed a data-mining approach for handling an imbalanced transaction dataset. The method increases the representation of the minority, fraudulent class through synthetic oversampling and reduces the majority, legitimate class through simple undersampling, while taking care not to remove so many samples that the overall characteristics of the data are lost. Several classification methods were then evaluated, and k-nearest neighbours produced the best result in the reported experiment. The results indicate that the proposed method achieved a better recall-based outcome than the other data-mining methods considered for this imbalanced dataset. The related literature also suggests that combining methods can improve performance across different measures, such as accuracy and computational cost [3, 5, 8].

References

[1] Nitesh V. Chawla, Aleksandar Lazarevic, Lawrence O. Hall, and Kevin W. Bowyer (2003). “SMOTEBoost: Improving Prediction of the Minority Class in Boosting.” Lecture Notes in Computer Science, vol. 2838, pp. 107–119.

[2] M. Farahmand, M. Tafahomi, and A. Elhami. “A Method for Classification and Detection of Bank Transactions Using Imbalanced Datasets.” 1381 (2002/2003). (The original Persian reference does not provide complete publication details; author names/title should be checked against the cited source.)

[3] Wei Wei, Jinjiu Li, Longbing Cao, Yuming Ou, and Jiahang Chen. “Effective Detection of Sophisticated Online Banking Fraud on Extremely Imbalanced Data.” World Wide Web, vol. 16, no. 4, 2013, pp. 449–475. https://doi.org/10.1007/s11280-012-0178-0.

[4] S. McAlearney. “TJX Data Breach: Ignore Cost Lessons and Weep.” CIO, August 7, 2008.

[5] Manoel Fernando Alonso Gadi, Xidi Wang, and Alair Pereira do Lago. “Credit Card Fraud Detection with Artificial Immune System.” In Artificial Immune Systems (ICARIS 2008), Lecture Notes in Computer Science, vol. 5132, 2008, pp. 119–131. https://doi.org/10.1007/978-3-540-85072-4_11.

[6] Tharakunnel, Westland, and Bhattacharyya. “Data Mining for Credit Card Fraud: A Comparative Study.” 2010. (The author formatting in the original is unclear.)

[7] H. Farvaresh and M. Sepehri. “A Data Mining Framework for Detecting Fraud in Telecommunication.” 2010. (The original reference list appears to merge this entry with the next one.)

[8] E. W. T. Ngai, Yong Hu, Y. H. Wong, Yijun Chen, and Xin Sun. “The Application of Data Mining Techniques in Financial Fraud Detection: A Classification Framework and an Academic Review of Literature.” Decision Support Systems, vol. 50, no. 3, 2011, pp. 559–569.

[9] Linda Delamaire, Hussein Abdou, and John Pointon. “Credit Card Fraud and Detection Techniques: A Review.” Banks and Bank Systems, vol. 4, issue 2, 2009.

[10] Mehmet Aci, Cigdem İnan Aci, and Mutlu Avci. “A Hybrid Classification Method of k Nearest Neighbor, Bayesian Methods and Genetic Algorithm.” Expert Systems with Applications, vol. 37, no. 7, 2010, pp. 5061–5067. https://doi.org/10.1016/j.eswa.2009.12.004.

[11] Mehdi Esmaeili. “Data Mining Techniques and Concepts.” 1380 (2001/2002).